Varsity Companion

Product Requirement Documentation & 6-Month MVP Launch Strategy

Target Market: South Africa (Higher Ed) **Timeframe:** 6 Months to MVP **Initial Focus:** Information Science & IT **Status:** Execution Ready

1. Executive Summary & Vision

Varsity Companion is an AI-powered academic operating system built specifically for South African university students. It addresses the critical operational friction faced by tertiary students: severe academic overload, fragmented deadline tracking, high mobile data costs, intermittent connectivity, and poor visibility into bursaries, learnerships, and graduate career paths.

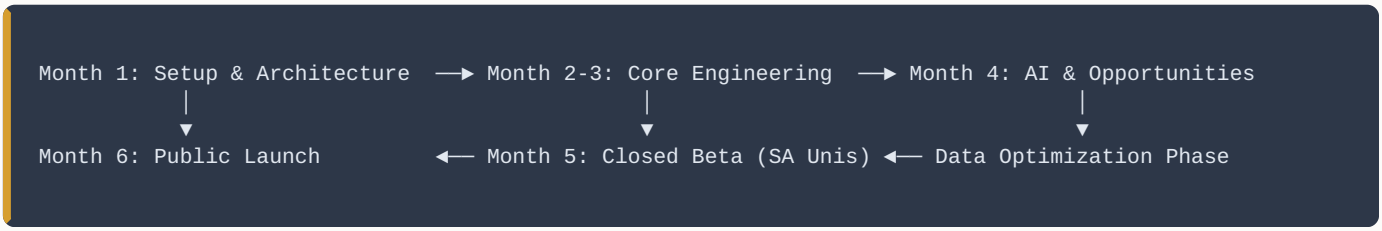
Core Value Proposition

A single, highly optimized mobile companion that reduces daily academic friction, provides localized study support, and connects South African students with real funding and employment opportunities—built ground-up for local infrastructure realities.

2. Problem & Solution Context

Local Student Challenges	Varsity Companion Technical Solution
Academic Fragmentation: Juggling separate LMS portals (Canvas/Moodle), unstructured WhatsApp groups, and personal calendars.	**Unified Academic Tracker:** Centralized task, module, and exam logging with intelligent priority sorting and local reminders.
High Data Costs & Connectivity: Inability to stream high-bandwidth video or stay continuously online.	**Offline-First Architecture:** Full offline local database (SQLite/ Room), lightweight JSON payloads, and text-first mode options.
Fragmented Opportunity Discovery: Missing critical funding (NSFAS, private bursaries) and internship closing dates.	**Targeted Opportunity Engine:** Matched bursary, learnership, and competition feed filterable by institution, field, and year.
Generic AI Tools: Tools like standard ChatGPT lack localized South African module context and low-data modes.	**Curated Local Assistant:** Topic-matched study resources and bandwidth-conscious YouTube video recommendations.

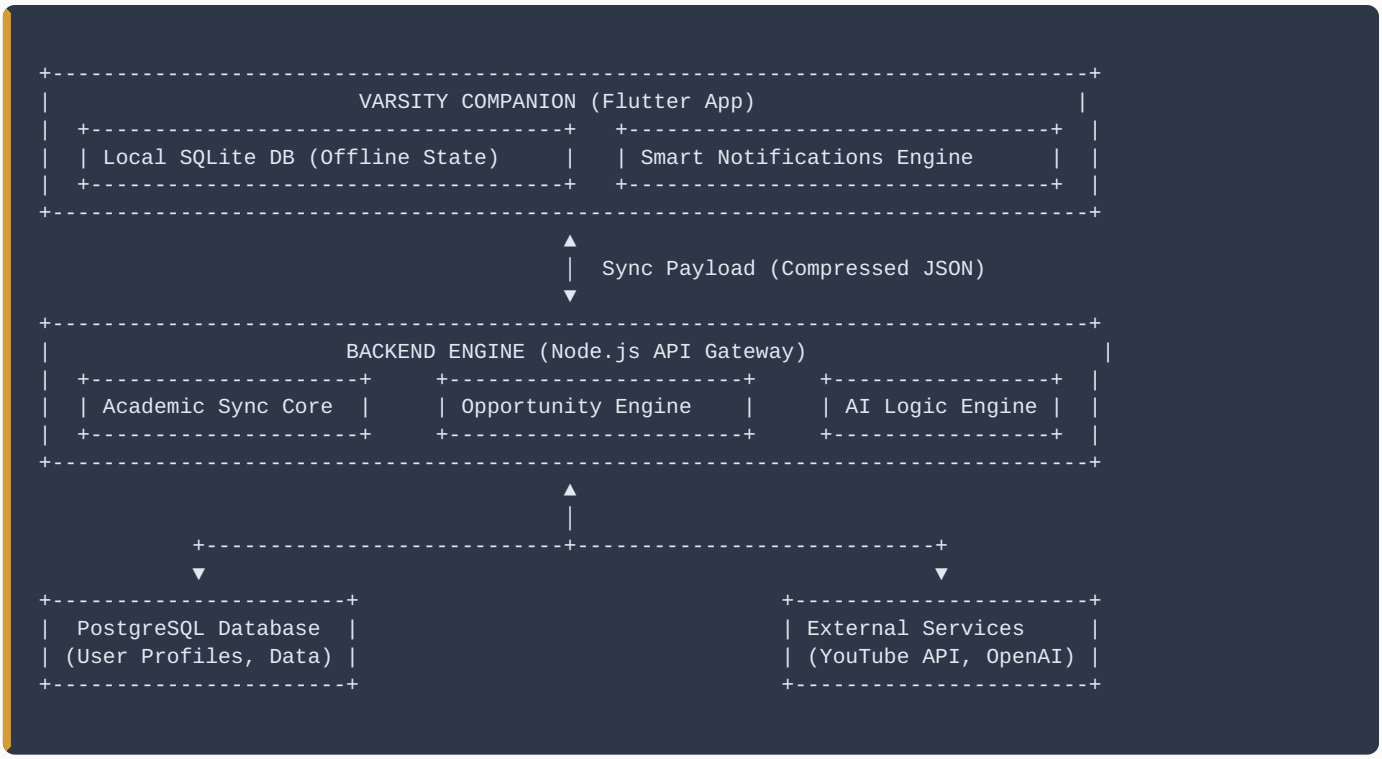
3. 6-Month MVP Roadmap & Key Milestones



Phase	Focus Area	Deliverables & Engineering Goals
Month 1	Foundation & Architecture	• Finalize technology stack (Flutter frontend, Node.js/Firebase backend, PostgreSQL/SQLite). • Map academic calendars and course structures for launch faculties (UP, Wits, UCT - IT/IS). • Design mobile-first wireframes optimized for low-bandwidth usage.
Month 2–3	Core Feature Engineering	• Implement local CRUD operations for modules, classes, and assignment deadlines. • Build offline-first synchronization engine with SQLite caching. • Develop native local notification engine for smart deadline reminders. • Construct structured opportunity feed (bursaries, learnerships) with search/filter mechanics.
Month 4	AI & Data Optimization	• Integrate YouTube API query wrappers for module-specific topic video curation (default 360p). • Deploy lightweight LLM endpoint for tagging assignments with key study concepts. • Build matching algorithm pairing student profile data (year, field, marks) to active bursary listings. • Implement payload compression (gzip/brotli) to enforce strict bandwidth limits (<15MB download).
Month 5	Closed Beta Testing	• Deploy closed beta (APK/TestFlight) to 100 IT/Information Science students across 2 SA universities. • Simulate poor connectivity (2G/3G throttling) to stress-test offline sync resilience. • Gather telemetry on daily active usage (DAU), task completion rates, and opportunity engagement.
Month 6	Public MVP Launch	• Launch Version 1.0 on Google Play Store (and Apple App Store). • Execute campus ambassador campaign across target IT student societies and WhatsApp networks. • Establish telemetry dashboards to prepare institutional and recruiter partnership reports.

4. Technical & System Architecture

The platform utilizes an **offline-first client design** to ensure uncompromised usability even during network loads or power outages.



5. Feature Matrix (MVP vs. Post-MVP Scope)

Module	In-Scope for 6-Month MVP	Scope Status
Academic Tracker	Manual module/task entry, local push notifications, priority weighting, countdown timers.	MVP CORE
LMS Integration	Direct automated sync with Canvas / Moodle / Blackboard via university credentials.	POST-MVP
Study Resources	Topic-matched YouTube video recommendations, concise concept summaries.	MVP CORE
AI Assessment	Full automated mock exam generation and note-parsing from custom PDF uploads.	POST-MVP
Opportunity Hub	Searchable database of verified bursaries, internships, learnerships with eligibility filtering.	MVP CORE
Application Portal	One-click direct job application submissions and live employer recruiting pipeline.	POST-MVP
Data & Offline	SQLite local caching, low-data text mode, compressed sync, zero video autoplay.	MVP CORE
WhatsApp Bot	Conversational WhatsApp interface for direct querying of deadlines and resources.	POST-MVP

6. Initial Go-To-Market & Growth Strategy

- **Hyper-Focused Launch Target:** Focus specifically on Information Science, Computer Science, and IT students at 2 major institutions (e.g., University of Pretoria, Wits University) to establish high initial network density.

- **Grassroots Distribution:** Partner with IT student societies, Developer Student Clubs (DSCs), and leverage class-level WhatsApp group networks via student ambassadors.
- **Value-Add Partnerships:** Aggregate anonymized demand insights to partner with corporate tech bursary providers, offering direct access to qualified graduate pools.